L2.6b More Optimization

§3.7 — when the interval has no ends

MONOTONIC FUNCTIONS

A purely increasing or purely decreasing function — one that either rises or falls over an interval — is called **monotonic**.

$$\boxed{f \text{ is monotonic on } I \iff f' \text{ has ONE sign throughout } I}$$

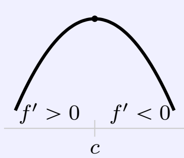
A critical number is where the sign of f' can change, so **the critical numbers are exactly where one monotonic interval ends and the next begins.** Split the domain at them and read off the sign of f' on each piece.
"$x^3 - 3x$ has two critical numbers, so three intervals — up, down, up"

The First Derivative Test for Absolute Extreme Values

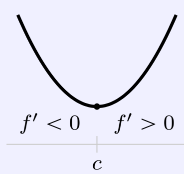
Let c be a critical number of a continuous f on an interval.

(a) $f' > 0$ for **all** $x < c$ and $f' < 0$ for **all** $x > c$ $\Longrightarrow$ $f(c)$ is the **absolute Maximum**

(b) $f' < 0$ for **all** $x < c$ and $f' > 0$ for **all** $x > c$ $\Longrightarrow$ $f(c)$ is the **absolute minimum**



absolute Max



absolute min

Why exactly one. With a second critical number, f can turn around again and drop below $f(c)$ further out. The hypothesis says **all** $x < c$ and **all** $x > c$, and a second critical number is precisely where that fails.

Two critical numbers on an open interval is not enough information — the theorem does not apply.

Which approach — closed or open. Solving an optimization problem requires either a closed interval or an open one, and the interval tells you which tool finishes it.

closed, $[a, b]$	the **Closed Interval Method**. Evaluate at every critical number *and* at both ends, then compare
open, (a, b) or $(0, \infty)$	the **First Derivative Test for Absolute Extreme Values**. There are no ends to evaluate, so the sign change across the critical number *is* the argument — and it needs exactly one

Local or absolute? The 2nd Derivative Test returns a **local** extremum — a claim about a neighbourhood of c and nothing beyond it. The theorem above returns an **absolute** one, over the whole interval, which is what an optimization question asked for.
"the cheapest can is the cheapest of ALL the cans, not the cheapest among nearby cans"

THE OPTIMIZATION TEMPLATE ON AN OPEN INTERVAL

① **Picture** · ② **Sort the question** — OEq, Constraint, Interval; eliminate every variable but one · ③ **Find critical pts** · ④ **Does this Maximize/min?** · ⑤ **Check endpoints**

On an open interval there are no ends, so there is nothing to evaluate at ⑤. The sign change carries the whole argument, and the theorem above is what names it.

"$h = 2r$ — where the answer is a relationship, the relationship is what gets boxed"

BUILDING THE OPTIMIZATION EQUATION

On the first day the question named the quantity — *area*, *volume*, *cost*. Here it names a *property*, and writing down what to optimize is the first real decision.

"closest" / "shortest"	minimize d^2, never d
"largest inscribed"	read the area off the picture — and **count the halves**: a rectangle $2x$ by $2y$ has area $4xy$, not xy
"how should it be cut"	the **sum** of the two pieces' areas, in one variable
"cheapest"	**cost**, not size — surfaces at different prices are different terms

Why d^2 is legal, and it is a theorem rather than a trick. $\sqrt{\ }$ is increasing on $[0, \infty)$, so it is monotonic there and preserves order: if $u < v$ then $\sqrt{u} < \sqrt{v}$. Whichever x makes d^2 smallest makes d smallest, **at the same** x.

"and the derivative stops being a chain rule on a radical"

WATCH OUT

✗ ~~minimize the distance $\sqrt{(x-a)^2 + (y-b)^2}$~~ → **minimize d^2. The square root is increasing, so the minimizer is the same and the derivative is a polynomial**

✗ ~~the rectangle inscribed in the circle has area xy~~ → **it is $(2x)(2y) = 4xy$ — the corner in the first quadrant is a quarter of the figure**

✗ ~~R is a letter, so differentiate it~~ → **name every constant before you differentiate. If R is fixed, $\frac{d}{dh}R^2 = 0$**

✗ ~~an endpoint answer is a mistake~~ → **it is a real answer — for 10 m of wire, the *maximum* area is at an end: put all of it into the square**

✗ ~~an open interval means you skip the justification~~ → **it means there are no ends to check, so the justification is the whole argument**